

Industrial AI Architect: Operations Report

Generated by LangGraph Multi-Agent Pipeline

Manufacturing Cell Operational Summary

Machine Utilization:

- * Total parts produced: 18,740
- * First-pass yield (FPY): 99.7%
- * Overall Equipment Effectiveness (OEE): 95.1% (exceeding the company target of 93%)

Connectivity Health:

* IoT Network Payloads:

- + Total payloads processed: 1200
- + Unique devices online: 5
- + Active workstations: WS04_ASSEMBLY_ROBOT, WS05_CMM_INSPECTION, WS01_CNC_MILLING, WS02_LATHE, WS03_HYDRAULIC_PRESS

* Sensor Anomalies:

- + Critical anomalies detected in the following sensors:
 - WS01_CNC_MILLING_WS01_CNC_MILLING_COOLANT_TEMP: Max 25.74 > Mean+3Std (25.42)
 - WS01_CNC_MILLING_WS01_CNC_MILLING_FEED_RATE: Max 362.05 > Mean+3Std (362.02)
 - WS01_CNC_MILLING_WS01_CNC_MILLING_SPINDLE_SPEED: Max 4552.14 > Mean+3Std (4543.07)
 - WS02_LATHE_WS02_LATHE_CUTTING_FORCE: Max 649.20 > Mean+3Std (645.45)
 - WS02_LATHE_WS02_LATHE_MOTOR_CURRENT: Max 15.77 > Mean+3Std (15.63)

- WS02_LATHE_WS02_LATHE_SPINDLE_SPEED: Max 2868.49 > Mean+3Std (2861.61)
- WS02_LATHE_WS02_LATHE_VIBRATION: Max 0.20 > Mean+3Std (0.20)
- WS03_HYDRAULIC_PRESS_WS03_HYDRAULIC_PRESS_CYCLE_TIME: Max 4.37 > Mean+3Std (4.35)
- WS03_HYDRAULIC_PRESS_WS03_HYDRAULIC_PRESS_HYDRAULIC_PRS: Max 229.10 > Mean+3Std (227.48)
- WS04_ASSEMBLY_ROBOT_WS04_ASSEMBLY_ROBOT_CYCLE_TIME: Max 13.37 > Mean+3Std (13.26)
- WS04_ASSEMBLY_ROBOT_WS04_ASSEMBLY_ROBOT_JOINT_TORQUE: Max 40.85 > Mean+3Std (40.28)
- WS04_ASSEMBLY_ROBOT_WS04_ASSEMBLY_ROBOT_MOTOR_TEMP: Max 54.69 > Mean+3Std (54.61)
- WS05_CMM_INSPECTION_WS05_CMM_INSPECTION_AMBIENT_TEMP: Max 20.37 > Mean+3Std (20.31)
- WS05_CMM_INSPECTION_WS05_CMM_INSPECTION_CYCLE_TIME: Max 98.63 > Mean+3Std (98.55)
- WS05_CMM_INSPECTION_WS05_CMM_INSPECTION_HUMIDITY: Max 49.59 > Mean+3Std (49.55)
- WS05_CMM_INSPECTION_WS05_CMM_INSPECTION_PROBE_FORCE: Max 0.16 > Mean+3Std (0.16)

Immediate Maintenance Actions Required:

* Perform preventive maintenance on the following workstations:

- + WS01 - CNC Milling: Spindle oil change + filter replacement
- + WS02 - Lathe: Turret indexing accuracy verification
- + WS03 - Hydraulic Press: Pressure relief valve re-calibration

+ WS04 - Assembly Robot: TCP calibration + teach-point check

* Re-qualify the full probe system on WS05 - CMM Inspection

Knowledge Graph Status:

* Successfully updated Neo4j Knowledge Graph with connected Hierarchical Structure.

Pipeline Performance Summary:

- Data Ingestion Agent: 0.0318s | 97 tokens
- IoT Payload Agent: 0.0214s | 179 tokens
- PLC Log Agent: 0.0214s | 3446 tokens
- OCR Document Agent: 0.2087s | 253 tokens
- Sensor Data Agent: 0.2415s | 730 tokens
- Neo4j Knowledge Graph Agent: 3.1996s | 4333 tokens
- Industrial Analysis Agent: 213.8605s | 5475 tokens
- Industrial Analysis Agent: 86.0958s | 6373 tokens